Aditya Singh Shekhawat
Roll No.- 25BCON1494

DATE / /
PAGE No.

Tutorial Sheet - 7

PART - A

Q.1 Define a complex variable and complex function. Give one example.

=>

complex variable is of the form $z = x + iy$.
A complex function is a function of $z$ : $f(z) = u(x,y) + i\,v(x,y)$

E.g :- $f(z) = z^2 = (x+iy)^2$

Q.2 Define the limit of a complex function. How does it differ from the limit of a real function.

=> $\lim_{z \to 0} f(z) = l$

Difference from real function : Real → only 2 directions (left / right).
complex → infinite paths, so limit must be same along all.

Q.3 Define continuity of a complex function at a point $z = z_0$.

=> Function $f(z)$ is continuous at $z_0$ if : $\lim_{z \to 0} f(z) = f(z_0)$

Q.4 State the Cauchy Rieman equation in cartesian form.

=> $f(z) = u + iv$ exist first order partial derivation

$\frac{\partial u}{\partial x}$, $\frac{\partial u}{\partial y}$, $\frac{\partial v}{\partial x}$, $\frac{\partial v}{\partial y}$

Aditya Singh Shekhawat
Roll No.- 25BCON1494

DATE / /
PAGE No.

C-R equation :- $\frac{du}{dx} = \frac{dv}{dy}$ and $\frac{du}{dy} = \frac{-dv}{dx}$

Q. 5 Define analytic function. Give one example.

=> A function $f(z)$ is said to be analytic at a point $z = z_0$ if it is differentiable at the point $z_0$ and also at each point in some neighbourhood of the $z_0$.
Analytic function also known as regular function or holomorpic function $f(z) = z^2$

PART - B

Q. 1 Check whether the function $f(z) = z^2 + 3z + 2$ is analytic and find its derivative.

=> Fxnⁿ $f(z) = z^2 + 3z + 2$ is polynomial so it's analytic everywhere.
Derivative of $f(z)$

$f'(z) = 2z + 3$ Ans

Q.2 Verify the Cauchy - Riemann equations in cartesian form $f(z) = x^2 - y^2 + i(2xy)$ and hence find $df/dz$.

=> $f(z) = x^2 - y^2 + i(2xy)$

$u = x^2 - y^2$ $v = 2xy$

$\frac{du}{dx} = 2x$, $\frac{du}{dy} = -2y$, $\frac{dv}{dx} = 2y$, $\frac{dv}{dy} = 2x$

$\frac{du}{dx} = \frac{dv}{dy} \Rightarrow 2x = 2x$, and $\frac{du}{dy} = \frac{-dv}{dx} \Rightarrow 2y = 2y$

Function is analytic

Aditya Singh Shekhawat
Roll No.- 25BCON1494

DATE / /
PAGE No.

Derivative : $\frac{df}{dz} = \frac{du}{dx} + i\frac{dv}{dx} \Rightarrow 2x + i(2y) = 2z.$

Q.3 Show that $u(x,y) = x^2 - y^2$ is a harmonic function and find its harmonic conjugate.

$\Rightarrow$ $u(x,y) = x^2 - y^2$

$\frac{du}{dx} = 2x, \quad \frac{d^2u}{dx^2} = 2, \quad \frac{du}{dy} = -2y, \quad \frac{d^2u}{dy^2} = -2$

here $\frac{d^2u}{dx^2} + \frac{d^2u}{dy^2} = 2 + (-2) = 0$

Satisfies laplace equation

From C-R

$\frac{du}{dx} = 2x, \quad \frac{du}{dy} = -2y$

$v = -\int \frac{du}{dy}\, dx + \int \underbrace{\frac{du}{dx}}_{\text{independent of } x}\, dy + c$

$v = \int 2y\, dx + \int 2x\, dy + c$

$v = 2xy + c$

PART - C

Q.1 Find the analytic function whose real part is $u = e^x(x\cos y - y\sin y)$.

$\Rightarrow$ $\phi_2(x,y) = \frac{du}{dy} = e^x(-x\sin y - 1.\sin y - y\cos y)$

$\phi_1(x,y) = \frac{du}{dx} = e^x(\cos y) + e^x(x\cos y - y\sin y$

Replacing $x$ by $z$ and $y$ by $0$

Aditya Singh Shekhawat
Roll No.- 25BCON1494



$\phi_1 = e^z(\cos 0) + e^z(z\cos 0 - 0\sin 0) = e^z(1+z)$

$\phi_2 = 0$

$f(z) = \int [\phi_1(z,0) - i\phi_2]\, dz + c$

$f(z) = \int [e^z(1+z) - i0]\, dz + c$

$= \int e^z(1+z)\, dz + c \Rightarrow \int e^z\, dz + \int e^z \cdot z\, dz + c$

$\Rightarrow e^z + z.e^z - e^z + c \Rightarrow z.e^z + c$ Ans

Q.2 If $u(x,y) = x^3 - 3xy^2$, show that $u$ is harmonic and find its harmonic conjugate.

$\Rightarrow$ $u = x^3 - 3xy^2$

$\frac{du}{dx} = 3x^2 - 3y^2$, $\frac{du}{dy} = -6xy$, $\frac{d^2u}{dx^2} = 6x$, $\frac{d^2u}{dy^2} = -6x$

Laplace eqn :- $\frac{d^2u}{dx^2} + \frac{d^2u}{dy^2} = 0 \Rightarrow 6x - 6x = 0$

Satisfied, Hence Harmonic

Harmonic conjugate $\Rightarrow$ $v = -\int \frac{du}{dy}\, dx + \int \frac{du}{dx}\, dy + c$

$\Rightarrow \int 6xy\, dx + \int -3y^2\, dy + c \Rightarrow 6y\int x\, dx + (-3)\int y^2 dy + c$

$\Rightarrow 3yx^2 - y^3 + c$ Ans

Q.3 Prove that if $f(z)$ is analytic, then its real and imaginary parts are harmonic.

$\Rightarrow$ If $f(z) = u + iv$ is analytic then C-R equation holds true.

$\frac{du}{dx} = \frac{dv}{dy}$, $\frac{du}{dy} = \frac{-dv}{dx}$

Aditya Singh Shekhawat
Roll No.- 25BCON1494

DATE / /
PAGE No.

On differentiating again for u if $\frac{d^2u}{dx^2} + \frac{d^2u}{dy^2} = 0$

then it's harmonic

for v if on differentiating again $\frac{d^2v}{dx^2} + \frac{d^2v}{dy^2} = 0$

Then it's harmonic

Hence both real and imaginary parts are harmonic.